

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Monday, 24 August 2026, 8:59 PM**State** Finished**Completed on** Monday, 24 August 2026, 9:00 PM**Time taken** 49 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      long long arg1 = *(const long long *)a;
6      long long arg2 = *(const long long *)b;
7      if (arg1 < arg2) return -1;
8      if (arg1 > arg2) return 1;
9      return 0;
10 }
11
12 int main() {
13     int n;
14     if (scanf("%d", &n) != 1) return 0;
15
16     long long *arr = (long long *)malloc(n * sizeof(long long));
17     if (arr == NULL) return 0;
18
19     for (int i = 0; i < n; i++) {
20         if (scanf("%lld", &arr[i]) != 1) {
21             free(arr);
22             return 0;
23         }
24     }
25
26     qsort(arr, n, sizeof(long long), compare);
27
28     long long max_sum = 0;
29     for (int i = 0; i < n; i++) {
30         max_sum += arr[i] * i;
31     }
32
33     printf("%lld\n", max_sum);
34
35     free(arr);
36     return 0;
37 }
38

```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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